

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences across **3,900 transactions** to guide strategic decisions.



Dataset at a Glance

3,900

Purchases

18

Features

37

Missing Values

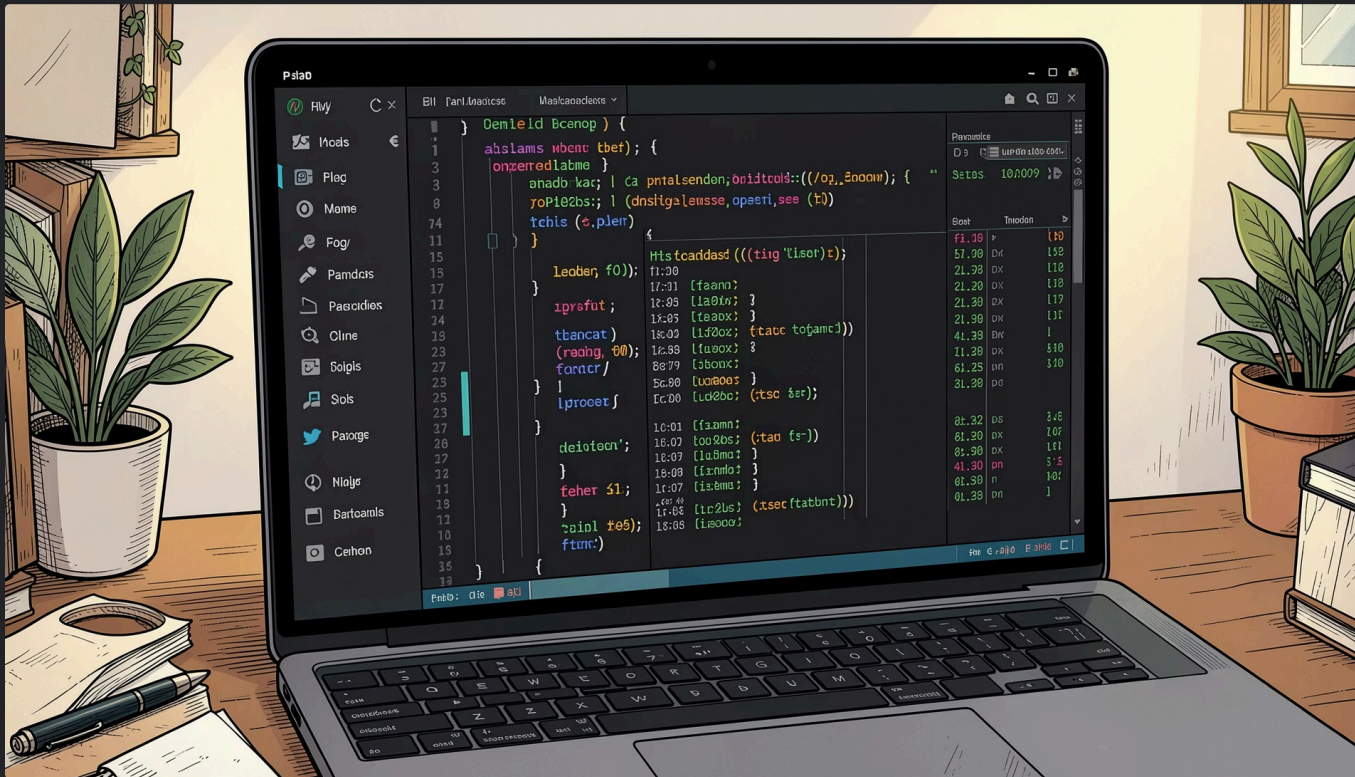
Review Rating column

\$59.76

Avg Purchase

Features span demographics, purchase details, shopping behavior, and subscription status.

Data Preparation in Python



→ Clean & Standardize

Imputed missing ratings by category median; renamed columns to snake_case

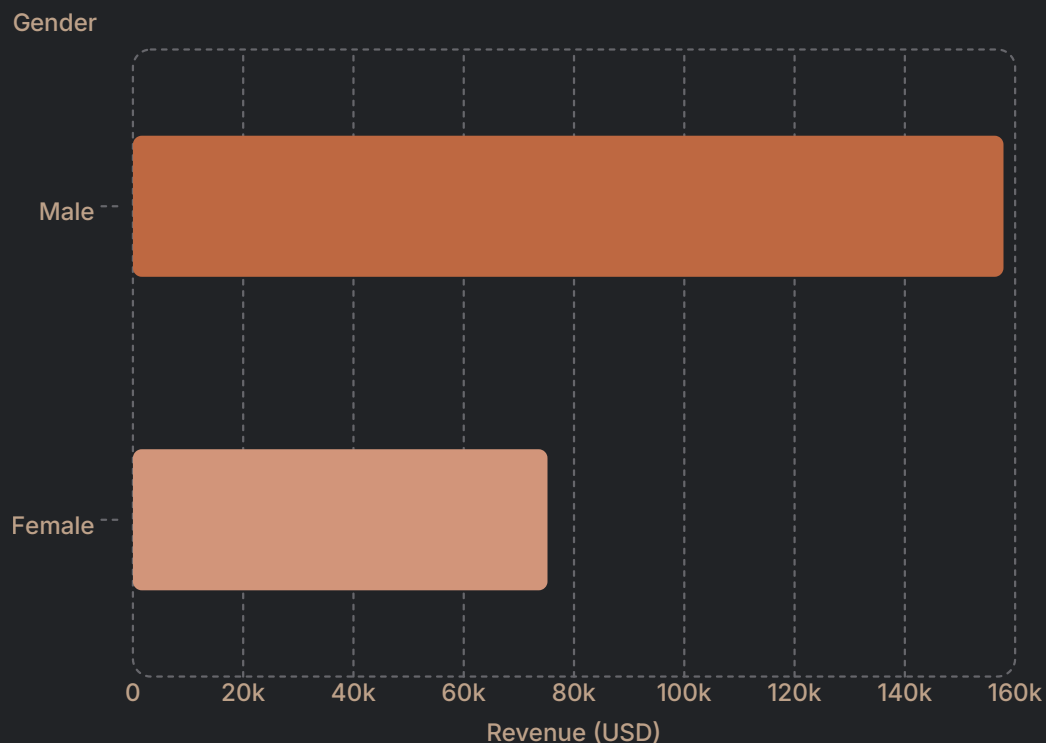
→ Feature Engineering

Created `age_group` bins and `purchase_frequency_days`

→ Load to PostgreSQL

Cleaned DataFrame loaded into database for SQL analysis

Revenue by Gender



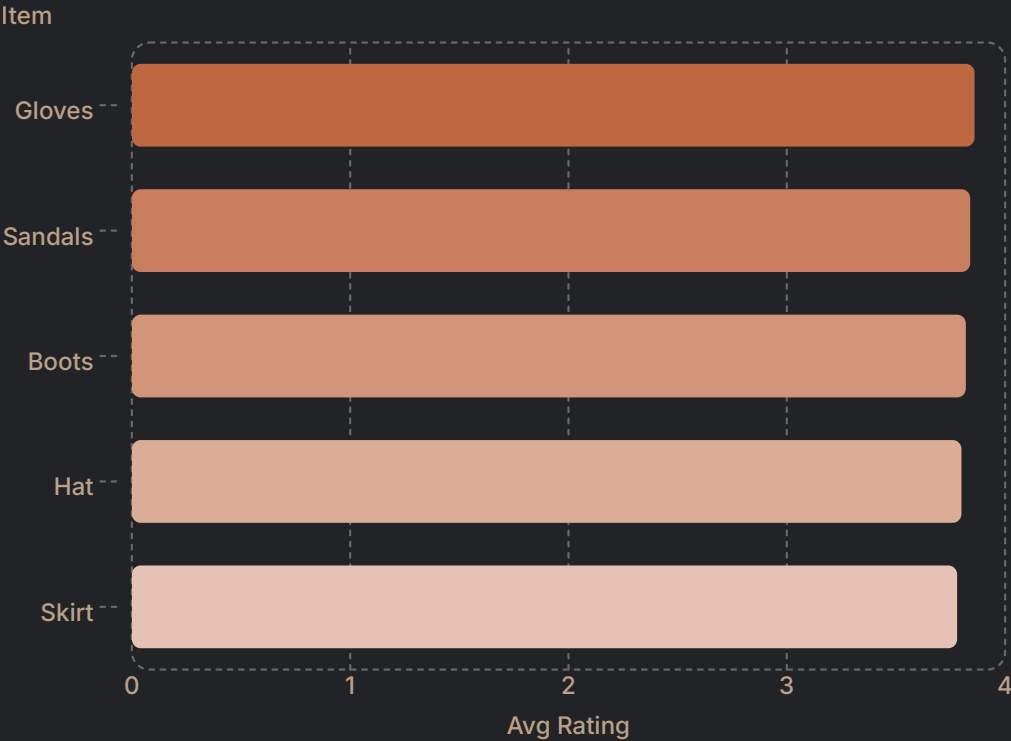
Male customers generate **2.1×** more revenue

Male: \$157,890 · Female: \$75,191

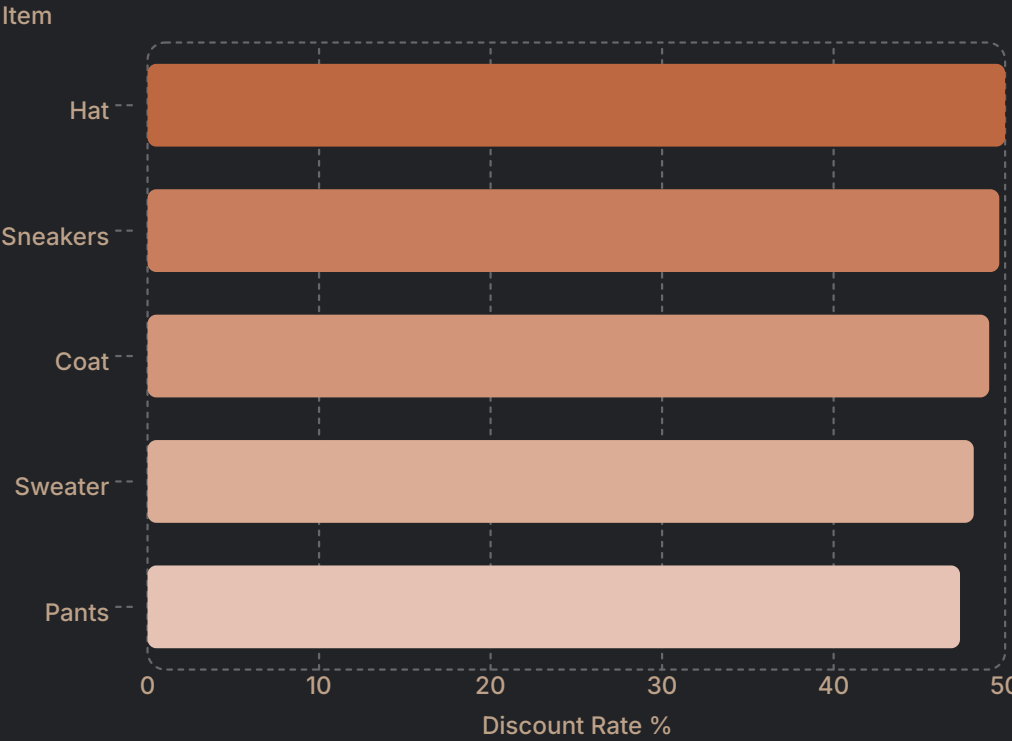
Male shoppers also represent **68%** of all transactions (2,652 of 3,900).

Top Products & Discount Behavior

Top 5 by Review Rating



Most Discount-Dependent



Shipping & Subscription Insights

Express Shipping

Avg spend: **\$60.48**
vs. Standard: \$58.46

Subscribers

1,053 customers
Avg spend: \$59.49

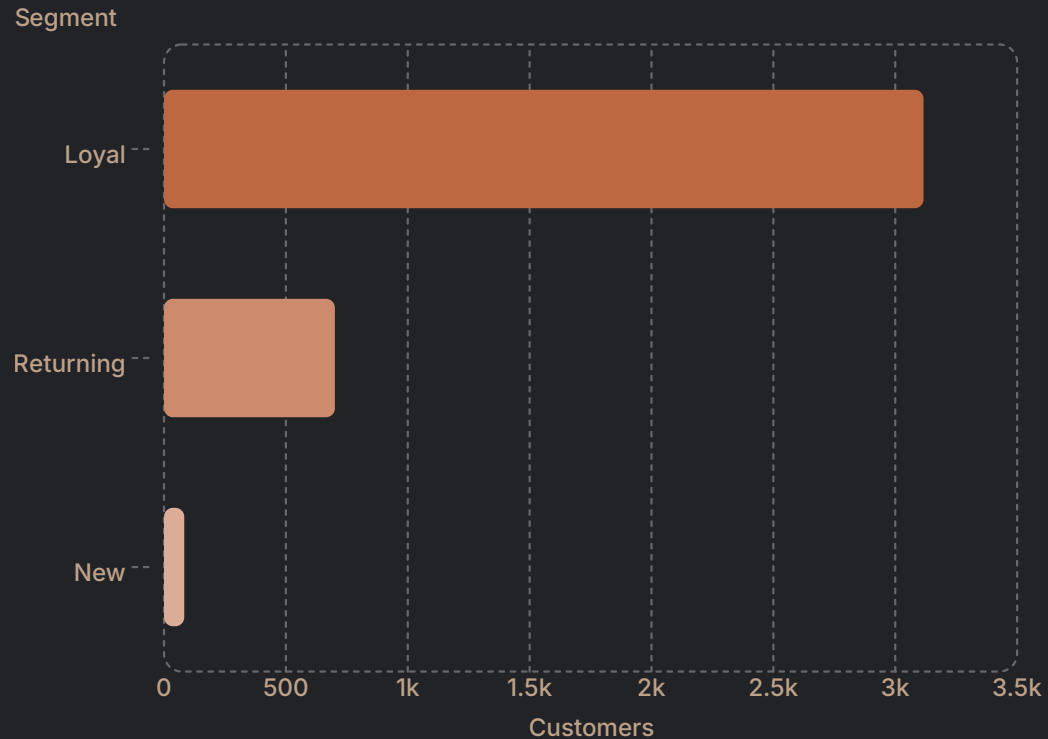
Non-Subscribers

2,847 customers
Avg spend: \$59.87

Repeat Buyers

958 subscribers have 5+ purchases vs. 2,518 non-subscribers

Customer Segmentation

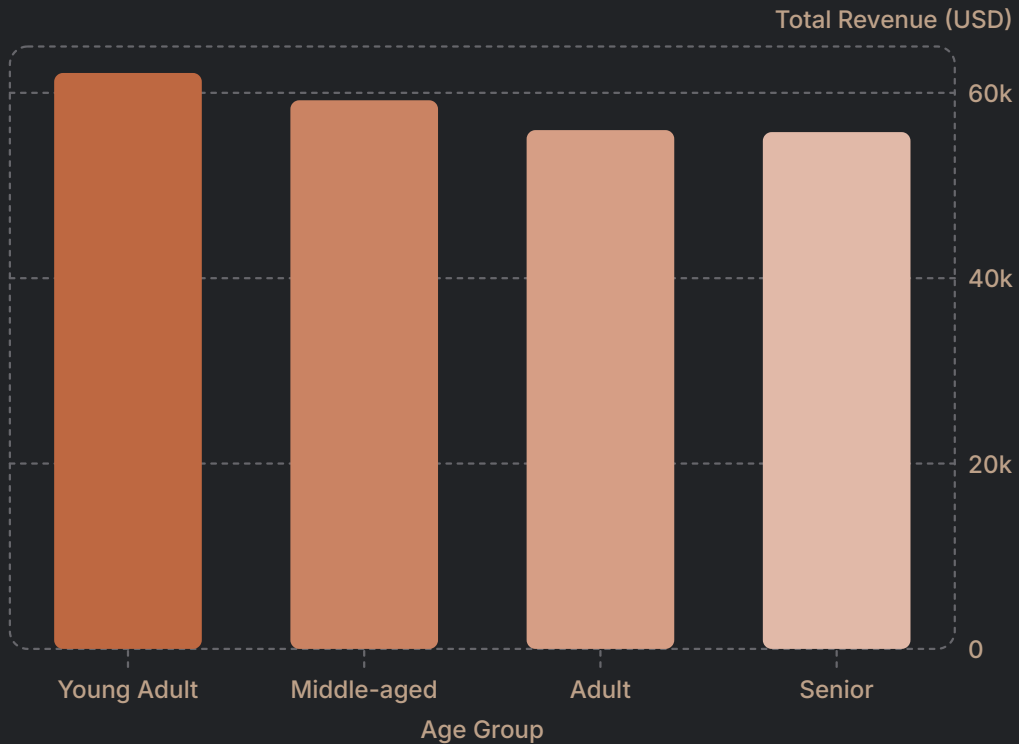


Loyal Customers Dominate

80% of customers are classified as Loyal based on purchase history.

Only **83 new customers** — opportunity for acquisition growth.

Revenue by Age Group



Young Adults Lead Revenue

Young Adult group contributes the highest total revenue at **\$62,143**.

Revenue is relatively balanced across all four age groups.

Power BI Dashboard



Interactive KPIs & Filters

3.9K Customers

Avg Rating: 3.75

27% Subscribers

Donut chart breakdown

Filters

Gender, Category, Shipping Type

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers into Loyal segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers

